Candidate Data Transformation Pipeline

Eightfold Engineering Internship Assignment – Technical Design

Design Philosophy: Build a *deterministic and explainable* transformation pipeline that prioritizes correctness, traceability, and configurable outputs over heuristic or probabilistic extraction.

1. Pipeline Overview

CSV + Resume + Recruiter Notes

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Source Adapters → Information Extraction → Normalization → Merge Engine

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Confidence + Provenance

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Projection (Runtime Config) → Validation → JSON Output

2. Canonical Schema & Normalization

Canonical Record (Internal Model)

- candidate_id
- full_name
- emails[]
- phones[]
- location
- links
- headline
- years_experience
- skills[]
- experience[]
- education[]
- salary_expectation
- availability
- provenance[]
- overall_confidence

Normalization Rules

- Emails → lowercase
- Phones → E.164
- Skills → canonical dictionary
- Dates → YYYY-MM
- Country → ISO 3166 Alpha-2

3. Merge / Conflict Resolution Policy

Field	Strategy
Full Name	Longest complete value
Emails	Normalize + Union
Phones	Normalize + Union (E.164)
Skills	Canonicalize + Union
Experience	Concatenate + Sort
Education	Concatenate + Deduplicate
Salary Expectation	Prefer Recruiter Notes
Availability	Prefer Recruiter Notes
Location	First non-empty value
Headline	First non-empty value

Candidate Matching

- Primary key: Normalized email
- Fallback: Normalized full name

Confidence Score

Computed deterministically using source reliability, successful normalization, validation status, and cross-source agreement.

4. Runtime Projection (Configurable Output)

The pipeline always produces a fixed internal canonical record. A runtime JSON configuration defines:

- Fields to include
- Field renaming / mapping
- Per-field normalization
- Confidence / provenance inclusion
- Missing-value policy (null / omit / error)

This cleanly separates the canonical model from the output schema without code changes.

Example (Custom Config)

```
{
  "fields": {
    "full_name": "name",
    "emails": "email_addresses",
    "phones": "phone_numbers",
    "skills": "skills",
    "experience": "work_history",
    "salary_expectation": "expected_ctc"
  },
  "include_confidence": true,
  "include_provenance": true,
  "on_missing": "null"
}
```

5. Edge Cases & Scope

Handled (Graceful Degradation)

- Missing/empty inputs (resume, notes, CSV)
- Duplicate emails / phones
- Invalid emails / phones
- Missing skills
- Missing education / experience
- Malformed input files
- Partial / incomplete data

Out of Scope

- OCR for scanned PDFs
- LLM / ML-based extraction
- Fuzzy entity resolution
- Non-English resumes
- Advanced semantic matching beyond simple identifiers

Design Goals

- Deterministic
- Explainable
- Robust
- Configurable
- Scalable